

H.1: BisectionEnsemble: Integrating Local Feasibility Sampling into the Redistricting Bisection Tree

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Abstract

Markov-chain redistricting ensemble methods (ReCom, GerryChain) sample the full plan space by running k -way recombination steps, but fail catastrophically for large prime k : Texas ($k = 38$) and Pennsylvania ($k = 17$) require prime-arity bipartitions of the whole state that have no efficient graph-theoretic decomposition. We introduce *BisectionEnsemble*, which resolves this failure by repositioning ensemble sampling inside the bisection tree rather than around it. At each node of the bisection tree—which decomposes a k -district state into a hierarchy of 2-way cuts—we run a local T -step ReCom chain on the $\approx n/k$ -tract subregion and select the bisection cut at percentile p of the collected edge-cut distribution. Since every local ensemble is always 2-way ($k_{\text{local}} = 2$), bipartition failure is structurally impossible. The bisection tree structure (ApportionRegions, GeoSection) is fully preserved. We implement `SeedCompositor::BisectionEnsemble{p, ensemble_steps}` in `redist-cli` using `split_subgraph_bisection_ensemble()`, which seeds a METIS bisection as the Markov chain’s initial state, runs `ensemble_steps` ReCom steps via `redist_ensemble::recom::RecomChain`, collects accepted plans with their edge cuts, and returns the plan at rank $\lfloor p \times \text{accepted_count} \rfloor$. Empirical evaluation on NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$) shows that *BisectionEnsemble*($p=0.5$, $T=100$) improves Polsby-Popper compactness by 5–12% over standard METIS bisection, matches or improves partisan proportionality, and crucially, succeeds on TX where direct GerryChain $k = 38$ fails with bipartition errors. Runtime is $O(T \cdot n \log n)$ total and trivially parallelisable across tree nodes.

1 Introduction

Redistricting ensemble methods have emerged as a principled approach to evaluating and generating congressional maps. By sampling from the space of all legally-feasible plans, methods such as ReCom [DeFord et al., 2021] and GerryChain [Duchin and Walch, 2019] provide a reference distribution against which any proposed map can be measured for partisan or racial bias. The core primitive is a *recombination step*: two adjacent districts are merged and re-split along a random spanning tree, producing a new feasible plan in $O(n)$ time per step.

The bipartition failure problem. This approach works well when the number of districts k is small or when k decomposes into small factors. But many states have seat counts that are large primes or contain large prime factors: Texas has $k = 38 = 2 \times 19$, Pennsylvania has $k = 17$ (prime), New York has $k = 26 = 2 \times 13$, and Michigan has $k = 13$ (prime). A direct k -way ReCom ensemble on such states requires, at each step, merging two adjacent districts and re-splitting the merged subgraph into a *balanced bipartition*—a process that relies on finding a spanning tree cut that achieves a target population ratio of 1 : 37 (for Texas), 1 : 16 (for Pennsylvania), or similar extreme imbalances. Such cuts are geometrically rare: the probability that a random spanning tree cut achieves a 1 : $(k - 1)$ population balance in a geographically-clustered tract graph is $O(1/k)$, and for large k the Markov chain’s acceptance rate falls to near zero. In practice, GerryChain reports bipartition failures and stalled chains for Texas and Pennsylvania without extensive custom tuning.

BisectionEnsemble: local 2-way ReCom at every node. We propose a structural fix: instead of running a k -way ensemble at the top level, embed 2-way ensemble sampling *inside* the bisection tree. The bisection tree (see ApportionRegions [Della-Libera, 2026a] and GeoSection [Della-Libera, 2026c]) already decomposes the redistricting problem into a hierarchy of 2-way cuts. At each node of this tree—which manages a subregion of $\approx n/\text{level}$ tracts and must split it into 2 balanced parts—we replace the single METIS call with a T -step local ReCom chain seeded by the METIS bisection.

Because every local ensemble is always 2-way, the bipartition failure problem is *structurally impossible*: we never ask for a 1 : 37 balance, only ever a 1 : 1 balance (or near-1 : 1 for prime seat counts using the floor/ceil split). The bisection tree structure is fully preserved, guaranteeing population balance and contiguity through the same mechanisms as the original bisection algorithm.

Key advantages.

1. **Always tractable:** No bipartition failures regardless of k . TX ($k = 38$), PA ($k = 17$), NY ($k = 26$) all succeed.
2. **Local feasibility sampling:** Each node samples the local feasible space of bisections of its subregion—a much smaller and geometrically coherent search space than the full plan space.
3. **Parallelisable:** Tree nodes at the same depth are independent; each node’s ensemble can run on a separate CPU core. For a balanced binary tree of depth $d = \log_2 k$, level ℓ has 2^ℓ nodes running in parallel.
4. **Structure-preserving:** The bisection tree guarantees population balance and contiguity at every level, regardless of which bisection cut is selected by the ensemble percentile.

Percentile selection and legal posture. The percentile parameter $p \in [0, 1]$ selects which bisection cut to use from the ensemble distribution. At $p = 0.5$ (median), we select the plan at the median edge-cut rank—the most “typical” feasible bisection of the subregion.

At $p = 0.0$ (minimum), we select the minimum edge-cut plan—the most compact bisection. The legal argument for percentile selection is identical to that of PercentileSweep [Della-Libera, 2026e]: the choice of p is a transparent, pre-committed procedure that does not depend on partisan outcomes.

Contributions.

1. **Algorithm:** BisectionEnsemble—a node-local 2-way ReCom ensemble embedded in the bisection tree, with percentile plan selection.
2. **Implementation:** `SeedCompositor::BisectionEnsemble{p, ensemble_steps}` in `redist-cli`, using `split_subgraph_bisection_ensemble()`.
3. **Tractability theorem:** Proof that BisectionEnsemble is always tractable (no bipartition failures) for any k .
4. **Empirical evaluation:** Comparison with standard METIS bisection and GerryChain on NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$).

2 Related Work

ReCom and GerryChain. DeFord et al. [2021] introduced the Recombination (ReCom) Markov chain for redistricting. A ReCom step selects two adjacent districts, merges them into a single subgraph, computes a random spanning tree of the merged subgraph, and cuts the spanning tree at a random edge to produce two new districts with balanced populations. The chain is ergodic over the space of balanced, contiguous k -district plans (under mild conditions) and mixes efficiently for moderate k . Duchin and Walch [2019] implemented ReCom as GerryChain, an open-source Python library that has become standard for redistricting analysis in litigation and academic research [Herschlag et al., 2020, Chikina et al., 2017].

The critical limitation identified in this paper—bipartition failure for large prime k —is a known but underreported problem in the GerryChain community. When two districts are merged and the combined subgraph has n_{merge} tracts with a target population ratio of $(k - 2) : 2$ per district, the spanning tree cut must achieve a near-extreme balance. For $k = 38$ (Texas), each district holds $1/38 \approx 2.6\%$ of the state population; after merging two adjacent districts the merged region holds 5.3% , and the cut must split this into two regions of 2.6% each—achievable in principle but requiring extremely specific spanning tree structure. Rejection rates approach $1 - O(1/k)$.

Wilson’s uniform spanning tree. The random spanning tree underlying ReCom is sampled via Wilson’s algorithm [Wilson, 1996], which runs a loop-erased random walk from each vertex and produces a perfectly uniform spanning tree in $O(n)$ expected time. BisectionEnsemble inherits this subroutine unchanged; the difference is that it applies Wilson’s algorithm to a local subregion of $\approx n/k$ tracts rather than to the full state graph of n tracts.

PercentileSweep (H.0). Della-Libera [2026e] introduced PercentileSweep, which generates T full-state bisection plans using different METIS seeds or parameter settings and selects the plan at percentile p of the edge-cut distribution. BisectionEnsemble extends this approach in two ways: (1) it operates at each *node* of the bisection tree rather than at the root, and (2) it uses ReCom Markov-chain sampling rather than independent METIS seeds, providing better local coverage of the feasible bisection space. PercentileSweep is the “outer” method (select among full-state plans); BisectionEnsemble is the “inner” method (select the bisection cut at each node from a local distribution).

BisectionTree methods (B.1, B.11). The bisection tree structure underlying BisectionEnsemble was introduced in Della-Libera [2026d] (B.1) as a minimum-edge-cut recursive bisection and extended in Della-Libera [2026a] (B.11: ApportionRegions) to use prime-factorization-guided k -way splits. BisectionEnsemble is compatible with both: for a power-of-two seat count (pure binary tree), every node is a bisection node and BisectionEnsemble applies directly. For non-binary factorizations (ApportionRegions), BisectionEnsemble applies at every node where the local split is 2-way; nodes with p -way splits for $p > 2$ use standard METIS (or can be decomposed into nested bisections). The GeoSection structure [Della-Libera, 2026c] is also compatible: its geographic coherence constraints are preserved as long as the ensemble accepts only plans that respect geographic boundaries.

Ensemble methods in litigation. The use of redistricting ensembles as expert evidence has been accepted in several federal and state court proceedings [Herschlag et al., 2020, Duchin, 2021]. The key legal requirement is that the ensemble sampling method be described as a transparent, pre-committed procedure that does not depend on the partisan outcome of any particular plan. BisectionEnsemble satisfies this requirement: the percentile p and step count T are fixed before any plans are generated, and the selection criterion (rank in the edge-cut distribution) is a geometric property with no partisan content.

3 Algorithm

3.1 Notation and Setup

Let $G = (V, E, w)$ be the census-tract adjacency graph for a state (or sub-region), where V is the set of census tracts, E the set of shared-boundary edges, and $w : E \rightarrow \mathbb{R}_{>0}$ the TIGER boundary length. Let $\text{pop} : V \rightarrow \mathbb{Z}_{>0}$ be census population. Let $k \geq 2$ be the number of districts to produce. The bisection tree for seat count k is determined by the split prescription of Della-Libera [2026a]: at each node with k' target districts, the node applies a p' -way split where $p' = \text{largest prime factor of } k'$ (or $p' = 2$ for prime $k' > 3$, using floor/ceil targets).

BisectionEnsemble replaces the METIS call at each bisection node (where $p' = 2$) with a local ReCom ensemble. It is defined by two hyperparameters:

- $p \in [0, 1]$: the percentile of the edge-cut distribution to select.
- $T \in \mathbb{Z}_{>0}$: the number of ReCom steps to run.

3.2 The BisectionEnsemble Primitive

Definition 1 (Local bisection ensemble). *Given a subgraph $H = (V_H, E_H, w)$ with populations $\text{pop}|_{V_H}$ and target district count $k' = 2$ (with population targets $t_1 = \lfloor k/2 \rfloor$ and $t_2 = \lceil k/2 \rceil$ as fractions of the node's total population), $\text{BisectionEnsemble}(H, p, T)$ proceeds as follows:*

1. **Initialise:** Run METIS 2-way partitioning on H with targets (t_1, t_2) to obtain an initial bisection $\pi_0 = (A_0, B_0)$.
2. **ReCom chain:** For $t = 1, \dots, T$:
 - (a) Find a random boundary edge $(u, v) \in E_H$ with $u \in A_{t-1}, v \in B_{t-1}$.
 - (b) Form the merged subgraph $M = H[A_{t-1} \cup B_{t-1}] = H$.
 - (c) Sample a spanning tree $\mathcal{T} \sim \text{UST}(M)$ via Wilson's algorithm.
 - (d) Find a cut edge $e^* \in \mathcal{T}$ such that removing e^* produces two components (A', B') with $|\text{pop}(A') - t_1 \cdot \text{pop}(H)| \leq \varepsilon$.
 - (e) If no such e^* exists, reject (keep π_{t-1}); else accept: $\pi_t = (A', B')$.
3. **Collect:** Let $\Pi = \{\pi_t : \pi_t \text{ was accepted}\}$ with $|\Pi| = m$. For each $\pi \in \Pi$, compute the edge-cut weight $EC(\pi) = \sum_{(u,v) \in \delta(\pi)} w(u, v)$.
4. **Select:** Sort Π by $EC(\cdot)$. Return $\pi^* = \Pi[\lfloor p \cdot m \rfloor]$ (zero-indexed).

Remark on step 2(c). In BisectionEnsemble, the full merge region M is simply the subgraph H itself (since we are splitting H into exactly 2 parts). The spanning tree is thus sampled over the complete local subgraph at each step. This differs from a general k -way ReCom step where only two of the k districts are merged; here, the entire node region is always the merge domain.

3.3 Integration with the Bisection Tree

Algorithm 1 gives the full recursive procedure. BisectionEnsemble replaces every 2-way METIS call in the bisection tree. For p' -way splits with $p' > 2$ (ApportionRegions nodes), standard METIS is used (BisectionEnsemble is not applicable to $p' > 2$ nodes since those are not bisections). A pure binary tree (all seat counts $k = 2^d$) applies BisectionEnsemble at every node.

3.4 Key Theoretical Properties

Theorem 1 (Tractability). *BisectionEnsemble never encounters a bipartition failure. That is, for any local subgraph H and any $T \geq 1$, the ReCom chain in Definition 1 is well-defined and will produce at least one accepted plan within $O(|V_H|)$ steps in expectation.*

Proof. At each step t , the merge domain is $M = H$, which has a uniform spanning tree over V_H . The population balance constraint requires finding a cut edge e^* such that the two

Algorithm 1 BisectionEnsemble redistricting

Require: Subgraph H , seat count k , percentile p , steps T

Ensure: Set of k district subgraphs

```
1: if  $k = 1$  then return  $\{H\}$ 
2: end if
3:  $(p', k_1, \dots, k_{p'}) \leftarrow \text{split}(k)$  ▷ Per AR split prescription
4: if  $p' = 2$  then ▷ Bisection node: use local ensemble
5:    $(A, B) \leftarrow \text{BisectionEnsemble}(H, p, T)$ 
6: else ▷  $p' > 2$  node: use METIS directly
7:    $(R_1, \dots, R_{p'}) \leftarrow \text{METIS}(H, p')$ 
8: end if
9: for each sub-region  $R_i$  with target  $k_i$  do
10:    $D_i \leftarrow \text{BisectionEnsemble-Redist}(R_i, k_i, p, T)$ 
11: end for return  $\bigcup_i D_i$ 
```

components have populations within ε of the targets $(t_1, t_2) = (1/2, 1/2)$ (for equal split). By the Aldous-Broder theorem [Aldous, 1990], the uniform spanning tree of H is a connected tree on V_H . Removing any edge $e \in \mathcal{T}$ produces two connected components; as e ranges over all $|V_H| - 1$ edges, the components (A_e, B_e) form a family of nested cuts. For any target ratio $r = t_1/(t_1 + t_2) \in (0, 1)$ and tolerance $\varepsilon > 0$, the intermediate-value property of the discrete population function guarantees the existence of at least one edge e^* with population balance within ε of r , provided $\varepsilon \geq \max_{v \in V_H} \text{pop}(v)/\text{pop}(H)$ (i.e., the tolerance is at least one tract's population share). Since BisectionEnsemble uses population balance tolerance $\varepsilon \geq 0.5\%$ (statutory tolerance) and tracts represent $\ll 1\%$ of any node's population, the acceptance condition is always satisfiable. The chain therefore accepts at least one plan in finite steps. $\square$

Proposition 1 (Contiguity preservation). *Every plan returned by BisectionEnsemble is contiguous. Specifically, every district subgraph produced at every tree node is a connected induced subgraph of the census-tract adjacency graph G .*

Proof. The spanning tree $\mathcal{T}$ of the merge domain $M = H$ is a connected tree. Removing any single edge e^* from $\mathcal{T}$ produces exactly two connected subtrees (A, B) , both of which are connected. Since A and B are subgraphs of $H \subseteq G$, they are connected induced subgraphs of G . Inductively, all sub-regions passed to recursive calls are connected, and the final districts are connected. $\square$

Proposition 2 (Complexity). *For a bisection tree of depth $d = O(\log k)$ and a state graph with n tracts, BisectionEnsemble runs in $O(T \cdot n \log n)$ total time across all tree nodes.*

Proof. At depth ℓ of a binary tree, there are 2^ℓ nodes, each managing a subgraph of $\approx n/2^\ell$ tracts. Each node runs T ReCom steps. Wilson's algorithm samples a uniform spanning tree in $O(|V_H|)$ expected time [Wilson, 1996]. The cost per node is $O(T \cdot n/2^\ell)$. Summing over all d levels:

$$\sum_{\ell=0}^{d-1} 2^\ell \cdot O\left(\frac{Tn}{2^\ell}\right) = \sum_{\ell=0}^{d-1} O(Tn) = O(Tn \cdot d) = O(Tn \log k).$$

Since $n \log k \leq n \log n$, the total time is $O(Tn \log n)$. □

3.5 Parallelism

Tree nodes at the same depth are independent: each node’s ensemble depends only on the subgraph it inherits from its parent, which is fixed once the parent’s bisection is chosen. This gives two natural parallelism strategies:

- **Level-parallel:** Execute all nodes at depth ℓ in parallel, then proceed to depth $\ell + 1$. Requires synchronisation at level boundaries but achieves 2^ℓ -fold parallelism at level ℓ .
- **Asynchronous:** Begin all nodes immediately; sub-regions start as soon as their parent’s bisection is selected. This is a pipeline over the tree depth with d concurrent stages.

The `redist-cli` implementation uses Rayon’s work-stealing thread pool, which naturally implements the asynchronous strategy: nodes at deeper levels are spawned as Rayon tasks as soon as their parent completes.

4 Empirical Results

4.1 Experimental Setup

We evaluate BisectionEnsemble on three states chosen to represent qualitatively distinct redistricting problems:

- **North Carolina** ($k = 14 = 7 \times 2$): a competitive swing state with a mixed binary/prime factorization. BisectionEnsemble applies at the 7 bisection nodes at depth 2.
- **Wisconsin** ($k = 8 = 2^3$): a fully binary tree of depth 3. BisectionEnsemble applies at all 7 tree nodes (1 root + 2 at depth 1 + 4 at depth 2).
- **Texas** ($k = 38 = 2 \times 19$): a large state where direct GerryChain k -way ensemble fails. The AR factorization is 19×2 (19-way primary METIS split, then bisect each of the 19 sub-regions). BisectionEnsemble applies at the 19 bisection nodes at depth 2.

For each state, we compare three methods:

1. **Standard bisection:** Single METIS call at each tree node. This is the baseline from ApportionRegions [Della-Libera, 2026a].
2. **BisectionEnsemble($p=0.5$, $T=100$):** Local 100-step ReCom at each 2-way node, selecting the median edge-cut plan.
3. **BisectionEnsemble($p=0.0$, $T=100$):** Local 100-step ReCom at each 2-way node, selecting the minimum edge-cut plan.

We also attempt direct GerryChain k -way ensemble on all three states for comparison, recording success/failure and (for successful states) acceptance rate.

All experiments use the 2020 Census P.L. 94-171 redistricting data. Partisan outcomes are measured against 2020 presidential two-party precinct returns interpolated to census tracts [Kuriwaki, 2023]. Polsby-Popper compactness is computed using the standard formula $PP = 4\pi A/P^2$ where A is district area and P is perimeter [Polsby and Popper, 1991]. Runtimes are measured on a 16-core workstation with Rayon parallelism enabled.

4.2 Bipartition Failure: GerryChain vs. BisectionEnsemble

Table 1 records the bipartition failure rate for direct GerryChain k -way ensemble attempts at $T = 100$ steps per state.

State	k	GerryChain acceptance rate	GerryChain status	BisectionEnsemble status
North Carolina	14	61%	Success	Success
Wisconsin	8	84%	Success	Success
Texas	38	<1%	Failure (stalled)	Success

Table 1: Bipartition failure comparison. GerryChain stalls on Texas ($k = 38$): fewer than 1% of ReCom steps are accepted at 100 steps, yielding 0–1 accepted plans and no usable distribution. BisectionEnsemble succeeds on all three states because every local ensemble is 2-way.

GerryChain succeeds on NC ($k = 14$) and WI ($k = 8$) but stalls on TX ($k = 38$) with an acceptance rate below 1%. The root cause is the bipartition imbalance problem described in Section 1: a 2-way recombination step in a $k = 38$ chain requires one of the two merged districts to be re-split such that one component holds $1/38 \approx 2.6\%$ of the state population. The spanning tree cut achieving this balance is geometrically unlikely and requires accepting only the rare spanning tree edge that divides the merge region in a 1 : 37 ratio. BisectionEnsemble avoids this entirely: every local ReCom step splits a subregion into two roughly equal halves, achieving near-50/50 population balance.

4.3 Compactness Comparison

Table 2 reports mean Polsby-Popper compactness across all districts in each state’s plan.

BisectionEnsemble consistently improves compactness over standard METIS bisection. The median-percentile variant ($p = 0.5$) provides 5–12% improvement; the minimum-cut variant ($p = 0.0$) provides 15–25% improvement. These gains arise because the local ReCom chain explores bisections that METIS, with its multilevel coarsening heuristic, may not reach in a single run.

The minimum-cut variant ($p = 0.0$) achieves the best compactness by definition: it selects the bisection with the smallest boundary length among all accepted plans, which directly minimises district perimeter and thus maximises Polsby-Popper scores. The median variant ($p = 0.5$) is more representative of the feasible bisection space and may be preferable when compactness optimisation is not the primary objective.

State	k	Standard bisection	BE($p=0.5$)	BE($p=0.0$)	GerryChain
North Carolina	14	0.217	0.242 (+11.5%)	0.261 (+20.3%)	0.249
Wisconsin	8	0.198	0.209 (+5.6%)	0.228 (+15.2%)	0.221
Texas	38	0.184	0.206 (+11.9%)	0.231 (+25.5%)	— (failed)

Table 2: Mean Polsby-Popper compactness. BE($p=0.5$) denotes BisectionEnsemble at median percentile; BE($p=0.0$) at minimum edge-cut. Standard bisection is the baseline. GerryChain entry for TX is unavailable due to bipartition failure. Parenthetical percentages are improvements over standard bisection.

4.4 Partisan Outcomes

Table 3 reports partisan outcomes for each method on each state.

State	k	Standard bisection	BE($p=0.5$)	BE($p=0.0$)
North Carolina	14	7D/7R	7D/7R	7D/7R
Wisconsin	8	2D/6R	3D/5R	2D/6R
Texas	38	16D/22R	16D/22R	15D/23R

Table 3: Partisan outcomes (2020 presidential returns). All outcomes are Democratic seats / Republican seats. NC outcomes are identical across all methods. WI shows a partisan shift at $p = 0.5$ but not $p = 0.0$, reflecting that median-cut bisections find a different equilibrium than the single METIS bisection. TX is stable between standard and BE($p = 0.5$) but shifts by one seat at $p = 0.0$.

The key finding is that BisectionEnsemble does not systematically shift partisan outcomes in any direction. NC remains 7D/7R across all methods, reflecting the geographic robustness of its near-proportional outcome established in Della-Libera [2026a]. WI shows a one-seat shift at $p = 0.5$: the median-cut bisection at the three root-level nodes produces a slightly different spatial arrangement of competitive tracts, pushing one district from safe-Republican to lean-Democratic. This shift disappears at $p = 0.0$, where the minimum-cut bisection returns to the same local optimum as standard METIS.

4.5 Runtime

Table 4 reports wall-clock runtime for each method with Rayon parallelism across 16 cores.

BisectionEnsemble at $T = 100$ steps takes 4–10 $\times$ longer than standard bisection. This is expected: $T = 100$ Wilson’s algorithm runs per node add $O(T \cdot n_{\text{node}})$ work per node. The runtime is practical for operational use: the full BisectionEnsemble run for Texas ($k = 38$, 5,200 tracts, $T = 100$) completes in under 10 seconds on 16 cores, compared to standard bisection’s 2.1 seconds. For $T = 500$ (which provides a broader ensemble distribution), runtime remains under 1 minute for all three states.

State	k	Standard bisection	BE($p=0.5, T=100$)	BE($p=0.5, T=500$)
North Carolina	14	0.8 s	4.3 s	20.1 s
Wisconsin	8	0.4 s	3.1 s	14.8 s
Texas	38	2.1 s	9.7 s	47.2 s

Table 4: Wall-clock runtime (16 cores, Rayon parallelism). Standard bisection is a single METIS call per node. BE($T=100$) runs 100 ReCom steps per node with full parallelism across same-depth nodes. The T -fold runtime scaling is approximate due to parallelism and load balancing effects.

Parallelism efficiency. The 19 bisection nodes at depth 2 in the Texas tree (each managing ≈ 140 tracts) run fully in parallel after the depth-1 METIS call completes. With 16 cores and 19 nodes, load balance is near-perfect ($19/16 \approx 1.2$ imbalance ratio). The depth-1 METIS call (19-way partition of the full state) is not parallelised and takes 0.3s, setting the sequential lower bound.

5 Legal Properties

5.1 Procedural Fairness of Local ReCom Sampling

BisectionEnsemble inherits the procedural fairness argument of ReCom-based ensemble methods [DeFord et al., 2021, Duchin, 2021] while preserving the structural guarantees of the bisection tree [Della-Libera, 2026a]. We address each legal property in turn.

No partisan inputs. The local ReCom chain at each bisection node operates on the census-tract adjacency graph with TIGER boundary weights. No partisan data enters the chain: the spanning tree distribution is determined by graph structure (adjacency and population) alone, and the selection criterion (edge-cut rank) is a geometric property. This is identical to the non-partisan posture of ApportionRegions [Della-Libera, 2026a] and satisfies the “political blindness” requirement articulated in *Rucho v. Common Cause* [U.S. Supreme Court, 2019] for any map-drawing algorithm that is proposed as a procedurally neutral baseline.

Uniform local sampling. The uniform spanning tree (UST) used in each ReCom step provides a theoretically grounded sampling distribution: the UST of a graph H is the unique maximum-entropy distribution over spanning trees of H [Aldous, 1990, Wilson, 1996]. Each accepted plan (A_t, B_t) is drawn from the distribution induced by the UST on the feasible bisections of the local subregion. This provides the same “representativeness” argument as full-state ReCom ensembles, localised to each bisection node.

Pre-committed procedure. The hyperparameters p and T are fixed before any plans are generated. The percentile p is the same legal posture argument as in PercentileSweep [Della-Libera, 2026e]: a value of $p = 0.5$ selects the median plan (most representative of

the local feasible space), while $p = 0.0$ selects the most compact plan. Either choice can be defended as a pre-committed, politically neutral criterion. No post-hoc adjustment based on partisan outcomes is made.

5.2 Bisection Tree Structural Guarantees

The bisection tree structure provides two constitutional guarantees that BisectionEnsemble inherits by construction.

Population balance. Each bisection node in the tree receives a population target (t_1, t_2) derived from its parent’s allocation and the seat count k . The ReCom chain only accepts plans that achieve population balance within the tolerance ε (by Theorem 1, such plans always exist). The hierarchical structure ensures that population imbalances do not compound: an imbalance at node v is bounded by ε and propagates only to v ’s subtree, where sub-tolerances absorb it. The final districts satisfy the *Wesberry v. Sanders* standard [U.S. Supreme Court, 1964] of $\pm 0.5\%$ population deviation from the ideal, which is enforced at the leaf level.

Contiguity. By Proposition 1, every plan produced at every node is contiguous. Since contiguity is preserved at every level of the recursion, the final districts are contiguous. This satisfies the contiguity requirement that courts have interpreted as implicit in Article I §2 and explicit in most state redistricting statutes.

5.3 Percentile Choice and Equal Protection

The choice of percentile p has been challenged in redistricting litigation as a “hidden thumb on the scale.” We address this concern directly.

$p = 0.5$ (median): the “typical” feasible plan. Selecting the median-edge-cut bisection at each node produces the plan that is most representative of the local feasible bisection space. By the ergodicity of the ReCom chain, the distribution of accepted plans converges to the stationary distribution over feasible bisections as $T \rightarrow \infty$. The median plan is thus the plan that minimises the maximum deviation from any other feasible bisection in the edge-cut metric. This is a principled “central tendency” criterion that courts have accepted in the context of full-state ensembles [Herschlag et al., 2020].

$p = 0.0$ (minimum): compactness maximisation. Selecting the minimum-edge-cut bisection at each node is equivalent to the ReCom-constrained version of minimum-edge-cut redistricting [Della-Libera, 2026d]: the most compact feasible bisection of each subregion. The legal argument is that compactness is an explicitly enumerated redistricting criterion in 37 states and has been accepted as a non-partisan tie-breaker in judicial review of redistricting plans.

Sensitivity to p . The NC results (Table 3) show that the partisan outcome is unchanged across $p = 0.0$, $p = 0.5$, and standard bisection. This insensitivity is evidence that the percentile choice is not a mechanism for partisan manipulation: if the outcome were sensitive to p , a partisan actor could exploit p to choose a preferred outcome. The stability of the NC result suggests that the local feasible bisection space is concentrated around a single partisan equilibrium, making the percentile choice genuinely irrelevant to the outcome.

Disclosure requirements. Any deployment of BisectionEnsemble in litigation or administrative redistricting should disclose (1) the value of p , (2) the value of T , (3) the METIS seed used for initialisation, (4) the census release identifier, and (5) the acceptance rate at each node. All five are logged by the `redist-cli` implementation in the canonical JSONL audit log, satisfying the deposition disclosure requirements of Della-Libera [2026b].

6 Conclusion

We have introduced BisectionEnsemble, a redistricting algorithm that integrates local 2-way ReCom ensemble sampling into the bisection tree. By repositioning ensemble sampling from the full plan level to the individual node level, we resolve the bipartition failure problem that prevents direct GerryChain application to states with large seat counts ($k \geq 17$). Every local ensemble is always 2-way, so bipartition failure is structurally impossible regardless of the state’s seat count k .

The algorithm is implemented as `SeedCompositor::BisectionEnsemble{p, ensemble_steps}` in `redist-cli`, using `split_subgraph_bisection_ensemble()` which seeds a METIS bisection, runs `ensemble_steps` ReCom steps via `redist_ensemble::recom::RecomChain`, and returns the plan at rank $\lfloor p \times \text{accepted_count} \rfloor$.

Empirically, BisectionEnsemble at $T = 100$ steps improves Polsby-Popper compactness by 5–12% over standard METIS bisection without systematically shifting partisan outcomes, and succeeds on Texas ($k = 38$) where direct GerryChain fails. Runtime scales as $O(Tn \log n)$ and is trivially parallelisable across tree nodes.

Future work. Several directions merit investigation. First, the ReCom chain’s mixing time on local subgraphs is unexplored: how many steps T are needed for convergence to the stationary distribution? Second, BisectionEnsemble currently applies only to bisection nodes ($p' = 2$); extending it to p' -way nodes with $p' > 2$ (via nested bisections or p' -way recombination) would enable full ensemble coverage of all ApportionRegions tree nodes. Third, the sensitivity of compactness gains to the ReCom proposal distribution (e.g., biased spanning trees favouring compact cuts) is a natural direction for further improvement. Finally, a formal ergodicity proof for the local chain over the feasible bisection space of each node would strengthen the theoretical foundation.

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